Displaying Data from Multiple Tables Using Joins

1. Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the EMPNO, ENAME, SAL, DNAME and LOC in the output. Use a NATURAL JOIN to produce the results.

```
SQL> select empno, ename, sal, dname, loc from emp natural join dept;

EMPNO ENAME          SAL DNAME        LOC
----- ------------ ----- ------------ ------------
 7839 KING          5000 ACCOUNTING   NEW YORK
 7698 BLAKE         2850 SALES        CHICAGO
 7782 CLARK         2450 ACCOUNTING   NEW YORK
 7566 JONES         2975 RESEARCH     DALLAS
 7788 SCOTT         3000 RESEARCH     DALLAS
 7902 FORD          3000 RESEARCH     DALLAS
 7369 SMITH          800 RESEARCH     DALLAS
 7499 ALLEN         1600 SALES        CHICAGO
 7521 WARD          1250 SALES        CHICAGO
 7654 MARTIN        1250 SALES        CHICAGO
 7844 TURNER        1500 SALES        CHICAGO
 7876 ADAMS         1100 RESEARCH     DALLAS
 7900 JAMES          950 SALES        CHICAGO
 7934 MILLER        1300 ACCOUNTING   NEW YORK

14 rows selected.
```

2. The HR department needs a report of all employees. Write a query to display the JOB, MGR, SAL, COMM, DNAME of employees whose JOB is SALESMAN.

```
SQL> select job, mgr, sal, comm, dname from emp e, dept d where e.deptno=d.deptno and job='SALESMAN';

JOB            MGR   SAL   COMM DNAME
------------ ----- ----- ------ ------------
SALESMAN      7698  1600    300 SALES
SALESMAN      7698  1250    500 SALES
SALESMAN      7698  1250   1400 SALES
SALESMAN      7698  1500      0 SALES
```

3. The HR department needs a report of employees in LOC DALLAS. Display the ENAME, JOB, DEPTNO, and DNAME for all employees who work in DALLAS.

```
SQL> select ename, job, e.deptno, dname from emp e, dept d where e.deptno=d.deptno and loc='DALLAS';

ENAME        JOB          DEPTNO DNAME
------------ ------------ ------ ------------
JONES        MANAGER          20 RESEARCH
SCOTT        ANALYST          20 RESEARCH
FORD         ANALYST          20 RESEARCH
SMITH        CLERK            20 RESEARCH
ADAMS        CLERK            20 RESEARCH
```

4. Create a report to display employees' name and employee number along with their manager's name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.

```
SQL> select e.ename as employee, e.empno as "Emp#", m.ename as manager, m.empno as "Mgr#" from emp e, emp m where e.mgr=m.empno;

EMPLOYEE      Emp# MANAGER       Mgr#
------------ ----- ------------ -----
BLAKE         7698 KING          7839
CLARK         7782 KING          7839
JONES         7566 KING          7839
ALLEN         7499 BLAKE         7698
WARD          7521 BLAKE         7698
MARTIN        7654 BLAKE         7698
TURNER        7844 BLAKE         7698
JAMES         7900 BLAKE         7698
MILLER        7934 CLARK         7782
SCOTT         7788 JONES         7566
FORD          7902 JONES         7566
ADAMS         7876 SCOTT         7788
SMITH         7369 FORD          7902

13 rows selected.
```

5. Modify the previous query to display all employees including King, who has no manager. Order the results by the employee number.

```
SQL> select e.ename as employee, e.empno as "Emp#", m.ename as manager, m.empno as "Mgr#" from emp e left outer join emp m on e.mgr=m.empno orde
r by e.empno;

EMPLOYEE      Emp# MANAGER       Mgr#
------------ ----- ------------ -----
SMITH         7369 FORD          7902
ALLEN         7499 BLAKE         7698
WARD          7521 BLAKE         7698
JONES         7566 KING          7839
MARTIN        7654 BLAKE         7698
BLAKE         7698 KING          7839
CLARK         7782 KING          7839
SCOTT         7788 JONES         7566
KING          7839
TURNER        7844 BLAKE         7698
ADAMS         7876 SCOTT         7788
JAMES         7900 BLAKE         7698
FORD          7902 JONES         7566
MILLER        7934 CLARK         7782

14 rows selected.
```

6. The HR department needs a report on job grades and salaries. To familiarize yourself with the SALGRADE table, first show the structure of the SALGRADE table. Then create a query that displays the name, job, department name, salary, and grade for all employees.

```
SQL> create table salgrade (
  2  grade number,
  3  losal number,
  4  hisal number
  5  );

Table created.

SQL> insert into salgrade values (1,700,1200);

1 row created.

SQL> insert into salgrade values (2,1201,1400);

1 row created.

SQL> insert into salgrade values (3,1401,2000);

1 row created.

SQL> insert into salgrade values (4,2001,3000);

1 row created.

SQL> insert into salgrade values (5,3001,9999);

1 row created.

SQL> desc salgrade;
 Name                                                              Null?    Type
 ----------------------------------------------------------------- -------- --------------------------------------------
 GRADE                                                                      NUMBER
 LOSAL                                                                      NUMBER
 HISAL                                                                      NUMBER
```

```
SQL> select ename, job, dname, sal, grade from emp e, dept d, salgrade s where e.deptno=d.deptno and e.sal between s.losal and s.hisal;

ENAME      JOB        DNAME            SAL GRADE
---------- ---------- ------------ ----- -----
MILLER     CLERK      ACCOUNTING    1300     2
CLARK      MANAGER    ACCOUNTING    2450     4
KING       PRESIDENT  ACCOUNTING    5000     5
SMITH      CLERK      RESEARCH       800     1
ADAMS      CLERK      RESEARCH      1100     1
JONES      MANAGER    RESEARCH      2975     4
SCOTT      ANALYST    RESEARCH      3000     4
FORD       ANALYST    RESEARCH      3000     4
JAMES      CLERK      SALES          950     1
WARD       SALESMAN   SALES         1250     2
MARTIN     SALESMAN   SALES         1250     2
ALLEN      SALESMAN   SALES         1600     3
TURNER     SALESMAN   SALES         1500     3
BLAKE      MANAGER    SALES         2850     4

14 rows selected.
```

7. Display the ENAME, DNAME of all the employees. Also display those department name which do not have any employees working.

```
SQL> select ename, dname from emp right outer join dept on emp.deptno=dept.deptno;

ENAME      DNAME
---------- ------------
KING       ACCOUNTING
BLAKE      SALES
CLARK      ACCOUNTING
JONES      RESEARCH
SCOTT      RESEARCH
FORD       RESEARCH
SMITH      RESEARCH
ALLEN      SALES
WARD       SALES
MARTIN     SALES
TURNER     SALES
ADAMS      RESEARCH
JAMES      SALES
MILLER     ACCOUNTING
           OPERATIONS

15 rows selected.
```

8. Display the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.

```
SQL> select e.ename, e.hiredate, m.ename as manager, m.hiredate from emp e, emp m where e.mgr=m.empno and e.hiredate<m.hiredate;

ENAME      HIREDATE  MANAGER                                            HIREDATE
---------- --------- -------------------------------------------------- ---------
BLAKE      01-MAY-81 KING                                               17-NOV-81
CLARK      09-JUN-81 KING                                               17-NOV-81
JONES      02-APR-81 KING                                               17-NOV-81
ALLEN      20-FEB-81 BLAKE                                              01-MAY-81
WARD       22-FEB-81 BLAKE                                              01-MAY-81
ADAMS      23-MAY-87 SCOTT                                              13-JUL-87
SMITH      17-DEC-80 FORD                                               03-DEC-81

7 rows selected.
```

9. Display the EMPNO, ENAME, DNAME, LOC of those employees who are working as CLERK. Use the USING clause.

```
SQL> select empno, ename, dname, loc from emp join dept using(deptno) where job='CLERK';

     EMPNO ENAME      DNAME        LOC
---------- ---------- ------------ --------------------------------------------------
      7934 MILLER     ACCOUNTING   NEW YORK
      7369 SMITH      RESEARCH     DALLAS
      7876 ADAMS      RESEARCH     DALLAS
      7900 JAMES      SALES        CHICAGO
```

10. Display the ENAME, SAL, MGR, DNAME of employees whose salary is more than 2000. Use the ON clause.

```
SQL> select ename, sal, mgr, dname from emp join dept on emp.deptno=dept.deptno where sal>2000;

ENAME        SAL        MGR DNAME
---------- ----- ---------- ------------
KING        5000            ACCOUNTING
BLAKE       2850       7839 SALES
CLARK       2450       7839 ACCOUNTING
JONES       2975       7839 RESEARCH
SCOTT       3000       7566 RESEARCH
FORD        3000       7566 RESEARCH

6 rows selected.
```

11. Display the EMPNO, ENAME, JOB, DEPTNO, DNAME, LOC of employees. Use LEFT OUTER JOIN.

```
SQL> select empno, ename, job, emp.deptno, dname, loc from emp left outer join dept on emp.deptno=dept.deptno;

     EMPNO ENAME      JOB           DEPTNO DNAME        LOC
---------- ---------- --------- ---------- ------------ --------------------------------------------------
      7839 KING       PRESIDENT         10 ACCOUNTING   NEW YORK
      7782 CLARK      MANAGER           10 ACCOUNTING   NEW YORK
      7934 MILLER     CLERK             10 ACCOUNTING   NEW YORK
      7566 JONES      MANAGER           20 RESEARCH     DALLAS
      7788 SCOTT      ANALYST           20 RESEARCH     DALLAS
      7902 FORD       ANALYST           20 RESEARCH     DALLAS
      7369 SMITH      CLERK             20 RESEARCH     DALLAS
      7876 ADAMS      CLERK             20 RESEARCH     DALLAS
      7698 BLAKE      MANAGER           30 SALES        CHICAGO
      7499 ALLEN      SALESMAN          30 SALES        CHICAGO
      7521 WARD       SALESMAN          30 SALES        CHICAGO
      7654 MARTIN     SALESMAN          30 SALES        CHICAGO
      7844 TURNER     SALESMAN          30 SALES        CHICAGO
      7900 JAMES      CLERK             30 SALES        CHICAGO

14 rows selected.
```

12. **Display the ENAME, DNAME of employees. Use RIGHT OUTER JOIN.**

```
SQL> select ename, dname from emp right outer join dept on emp.deptno=dept.deptno;

ENAME      DNAME
---------- ------------
KING       ACCOUNTING
BLAKE      SALES
CLARK      ACCOUNTING
JONES      RESEARCH
SCOTT      RESEARCH
FORD       RESEARCH
SMITH      RESEARCH
ALLEN      SALES
WARD       SALES
MARTIN     SALES
TURNER     SALES
ADAMS      RESEARCH
JAMES      SALES
MILLER     ACCOUNTING
           OPERATIONS

15 rows selected.
```

13.Display the EMPNO, DNAME, LOC of employees. Use FULL OUTER JOIN.

```
SQL> select empno, dname, loc from emp full outer join dept on emp.deptno=dept.deptno;

     EMPNO DNAME        LOC
---------- ------------ -------------------------------------------------
      7839 ACCOUNTING   NEW YORK
      7698 SALES        CHICAGO
      7782 ACCOUNTING   NEW YORK
      7566 RESEARCH     DALLAS
      7788 RESEARCH     DALLAS
      7902 RESEARCH     DALLAS
      7369 RESEARCH     DALLAS
      7499 SALES        CHICAGO
      7521 SALES        CHICAGO
      7654 SALES        CHICAGO
      7844 SALES        CHICAGO
      7876 RESEARCH     DALLAS
      7900 SALES        CHICAGO
      7934 ACCOUNTING   NEW YORK
           OPERATIONS   BOSTON

15 rows selected.
```